



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

DIFFUSION MODELS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A AI/ML

TOTAL: 10 QUESTIONS

Q1 What is Diffusion Models and what AI/ML use case is it designed for?

Image Generation

Q6 How do you evaluate a model's performance in Diffusion Models?

Image Quality

Q2 What are the core abstractions in Diffusion Models?

Architecture

Q7 How do you deploy a model trained with Diffusion Models?

Deployment

Q3 How does Diffusion Models handle training or inference?

Training

Q8 How does Diffusion Models handle distributed or multi-GPU training?

Scalability

Q4 How does Diffusion Models manage data preprocessing?

Data Augmentation

Q9 How do you track experiments and versions in Diffusion Models?

Experiment Tracking

Q5 How does Diffusion Models support GPU/TPU acceleration?

Hardware Acceleration

Q10 What are common pitfalls when using Diffusion Models in production?

Model Drift



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Diffusion Models is a framework or service

Mitigation

Diffusion Models is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Diffusion Models organizes work around abstractions like

Initiatives

Diffusion Models organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Diffusion Models defines a model

Hardening

For training, Diffusion Models defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Diffusion Models provides data-parallel or model-parallel

Performance

Diffusion Models provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Diffusion Models provides data loaders, tokenizers, or

Gotchas

Diffusion Models provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights &

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Diffusion Models to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Diffusion Models detects available accelerators and moves

Optimization

Diffusion Models detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.